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10 CONCEPTUAL Q&A AI/ML · PRACTICE WORKSHEET

CONCEPTUAL SERIES
LSTM

Name: ______ Cohort/Batch: ______ Date: ______ Score: ______

LSTM PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A AI/ML

TOTAL: 10 QUESTIONS

Q1 What is LSTM and what AI/ML use case is it designed for? *Threat Model*

Q2 What are the core abstractions in LSTM? *Architecture*

Q3 How does LSTM handle training or inference? *Configuration*

Q4 How does LSTM manage data preprocessing? *Hardening*

Q5 How does LSTM support GPU/TPU acceleration? *SSO / IdP*

Q6 How do you evaluate a model's performance in LSTM? *Observability*

Q7 How do you deploy a model trained with LSTM? *Lifecycle*

Q8 How does LSTM handle distributed or multi-GPU training? *Compliance*

Q9 How do you track experiments and versions in LSTM? *Testing*

Q10 What are common pitfalls when using LSTM in production? *Tuning*

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 LSTM is a framework or service targeting

Mitigation

LSTM is a framework or service targeting a specific AI/ML domain training neural networks, building LLM pipelines, deploying models, or orchestrating agents. Understanding its scope helps you know when to use it vs alternatives.

A2 LSTM organizes work around abstractions like models,

Primitives

LSTM organizes work around abstractions like models, tensors, pipelines, agents, or embeddings. Learning these core types is the first step to using the framework effectively.

A3 For training, LSTM defines a model architecture,

Hardening

For training, LSTM defines a model architecture, a loss function, and an optimizer. For inference, a trained model processes input data and returns predictions. Batch processing and hardware acceleration are key performance levers.

A4 LSTM provides data loaders, tokenizers, or pipeline

Gotchas

LSTM provides data loaders, tokenizers, or pipeline components that transform raw data into the tensor or embedding format the model expects. Proper preprocessing is critical for model correctness and training speed.

A5 LSTM detects available accelerators and moves computation

Federation

LSTM detects available accelerators and moves computation to them. Specify the device explicitly (cuda, mps, tpu) to avoid accidentally running expensive operations on CPU. Mixed-precision (fp16/bf16) further reduces memory and increases throughput.

A6 Evaluation uses a held-out validation set and

Observability

Evaluation uses a held-out validation set and metrics (accuracy, F1, BLEU, perplexity) appropriate to the task. Monitor both training and validation metrics to detect overfitting. Use a test set only at the very end to report final results.

A7 Export the model to a portable format

Automation

Export the model to a portable format (ONNX, TorchScript, SavedModel). Serve it via a model server (TorchServe, TF Serving, Triton) or embed it in an API endpoint. Monitor inference latency and drift in production.

A8 LSTM provides data-parallel or model-parallel strategies to

Governance

LSTM provides data-parallel or model-parallel strategies to split work across GPUs. Data parallelism replicates the model on each GPU and averages gradients. Model parallelism splits layers across devices for models too large for one GPU.

A9 Use an experiment tracker (MLflow, Weights &

Assessment

Use an experiment tracker (MLflow, Weights & Biases, Comet) alongside LSTM to log hyperparameters, metrics, and artifacts. Track the code version and data version for full reproducibility.

A10 Common issues include training-serving skew (different

Optimization

Common issues include training-serving skew (different preprocessing), missing input validation, no latency SLA enforcement, models not versioned, and no process to retrain on drifted data distributions.